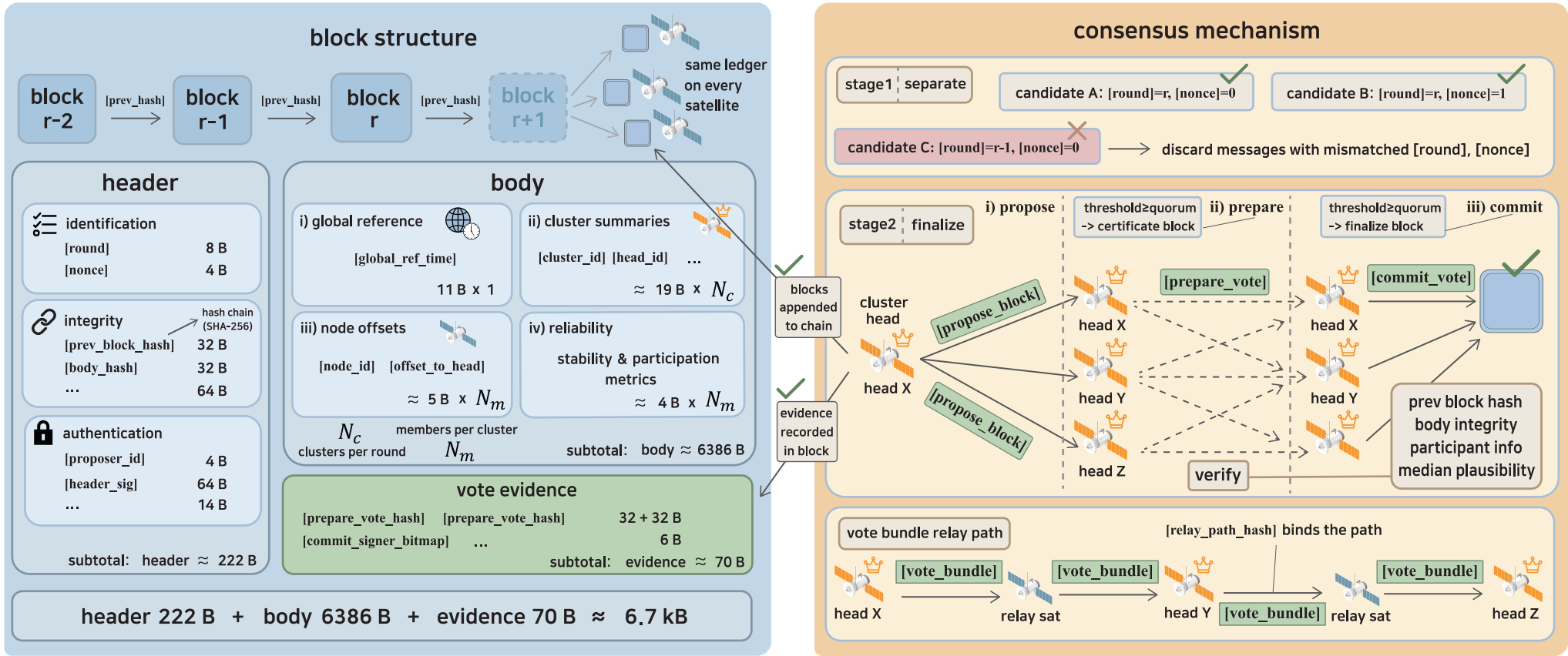

block structure
block r-2
[prev_hash]
block r-1
[prev_hash]
block r
[prev_hash]
block r+1
same ledger on every satellite
header
identification
[round] 8 B
[nonce] 4 B
integrity
hash chain (SHA-256)
[prev_block_hash] 32 B
[body_hash] 32 B
... 64 B
authentication
[proposer_id] 4 B
[header_sig] 64 B
... 14 B
subtotal: header ≈ 222 B
body
i) global reference
[global_ref_time]
11 B x 1
ii) cluster summaries
[cluster_id] [head_id] ...
≈ 19 B x N_c
iii) node offsets
[node_id] [offset_to_head]
≈ 5 B x N_m
iv) reliability
stability & participation metrics
≈ 4 B x N_m
N_c clusters per round
members per cluster N_m
subtotal: body ≈ 6386 B
vote evidence
[prepare_vote_hash] [prepare_vote_hash] 32 + 32 B
[commit_signer_bitmap] ... 6 B
subtotal: evidence ≈ 70 B
header 222 B + body 6386 B + evidence 70 B ≈ 6.7 kB
blocks appended to chain
evidence recorded in block
consensus mechanism
stage1 separate
candidate A: [round]=r, [nonce]=0
candidate B: [round]=r, [nonce]=1
candidate C: [round]=r-1, [nonce]=0
discard messages with mismatched [round], [nonce]
stage2 finalize
i) propose
threshold≥quorum -> certificate block
ii) prepare
threshold≥quorum -> finalize block
iii) commit
cluster head
head X
[propose_block]
[propose_block]
[prepare_vote]
[commit_vote]
head X
head Y
head Z
head X
head Y
verify
prev block hash
body integrity
participant info
median plausibility
vote bundle relay path
[relay_path_hash] binds the path
head X
[vote_bundle]
relay sat
[vote_bundle]
head Y
[vote_bundle]
relay sat
[vote_bundle]
head Z